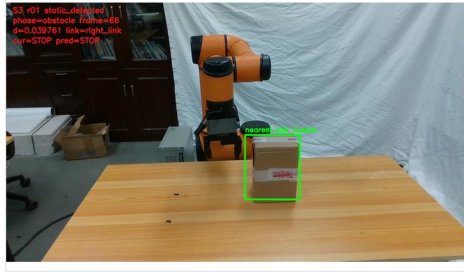


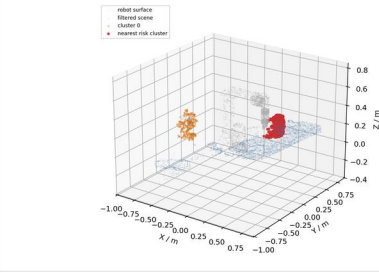
6.5.1 Real RGB-D perception snapshots: RGB, point cloud, and risk curves

RGB detection frame

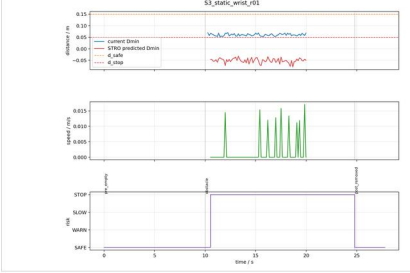
(a) Static obstacle near wrist/end: detected



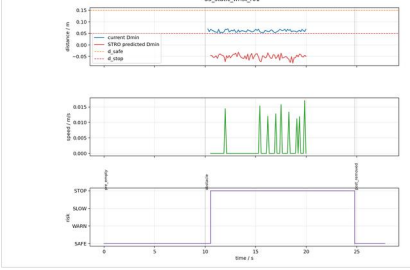
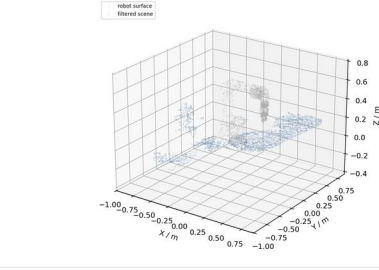
Filtered point cloud / clusters



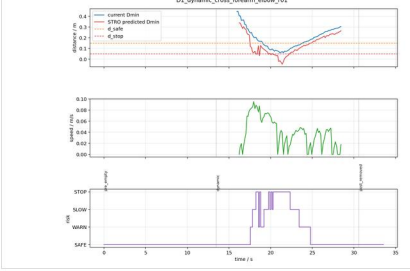
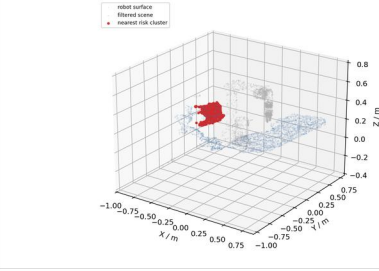
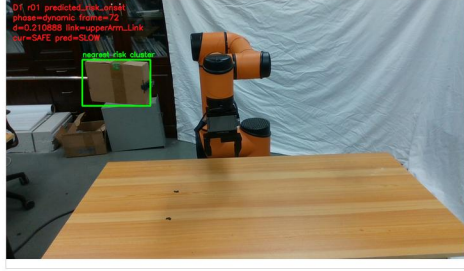
Distance-risk curve



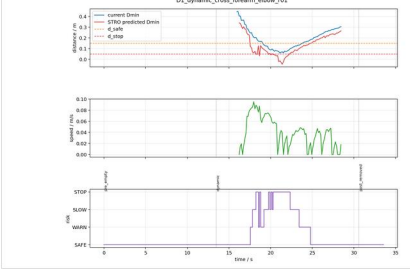
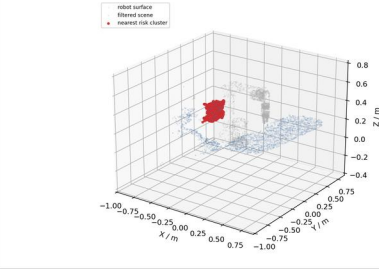
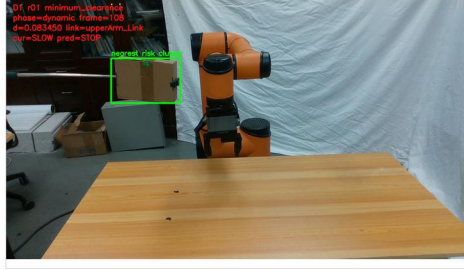
(b) Static obstacle removed: risk cleared



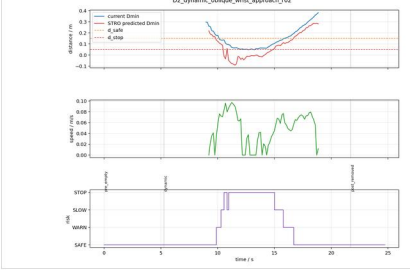
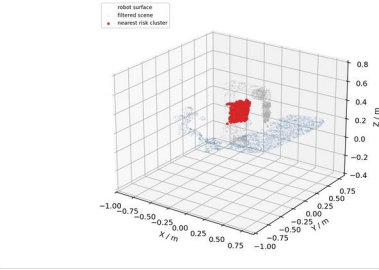
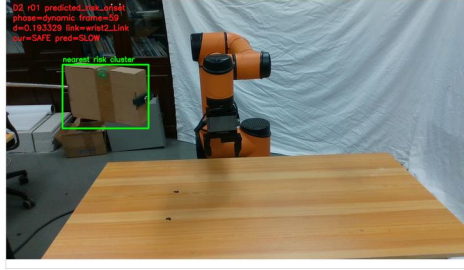
(c) D1 cross upper arm: predicted risk onset



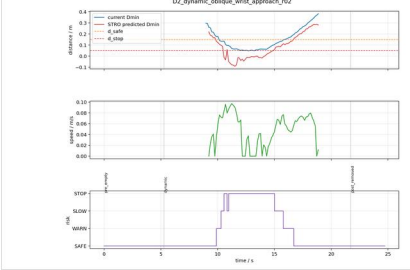
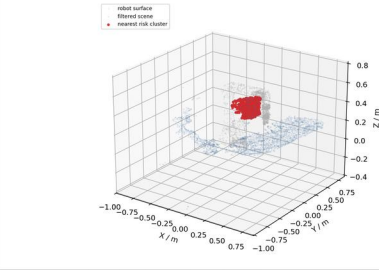
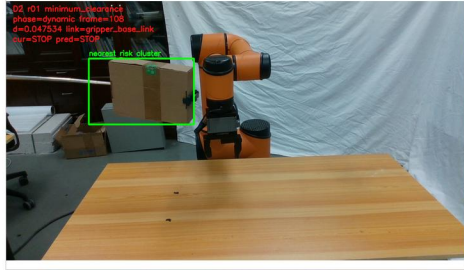
(d) D1 cross upper arm: minimum clearance



(e) D2 oblique wrist/end approach: predicted risk onset



(f) D2 oblique wrist/end approach: minimum clearance



Note: D2 curve uses the formal D2 r02 trial; D2 RGB/point-cloud frames are representative visual snapshots from the same scenario.